

Set-13: Geographic Data

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

Program:

```
# Install packages (Run once)
# install.packages("ggplot2")
# install.packages("maps")

library(ggplot2)
library(maps)

# Dataset
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200),
  Longitude = c(-100, -90, -80),
  Latitude = c(40, 35, 45)
)

# World map
world_map <- map_data("world")

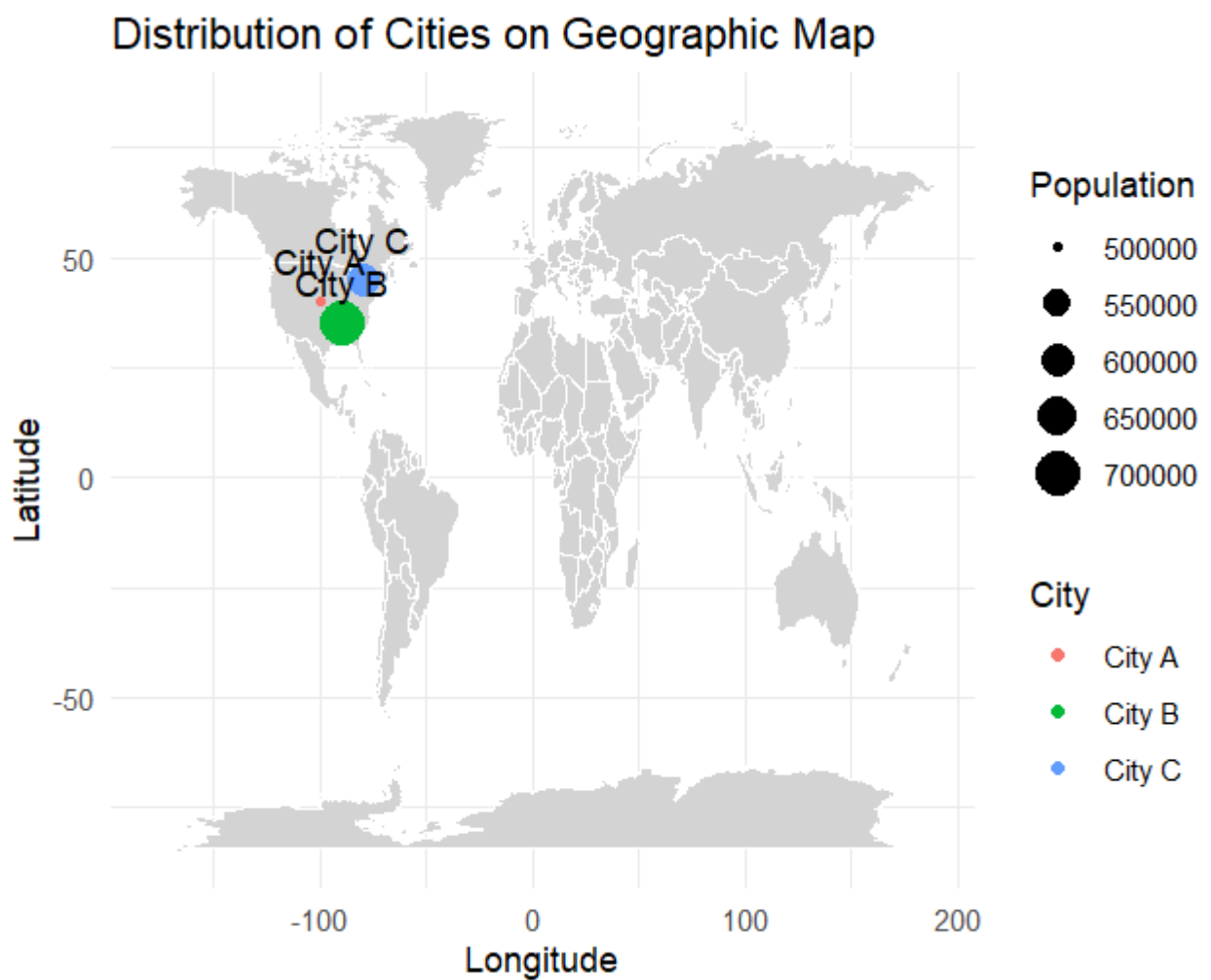
# Map Chart
ggplot() +
  geom_polygon(data = world_map,
    aes(x = long, y = lat, group = group),
    fill = "lightgray",
    color = "white") +
  geom_point(data = geo_data,
    aes(x = Longitude,
      y = Latitude,
      size = Population,
```

```

    color = City)) +
geom_text(data = geo_data,
    aes(x = Longitude,
    y = Latitude,
    label = City),
    vjust = -1) +
labs(title = "Distribution of Cities on Geographic Map",
    x = "Longitude",
    y = "Latitude",
    size = "Population",
    color = "City") +
theme_minimal()

```

Output:



2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

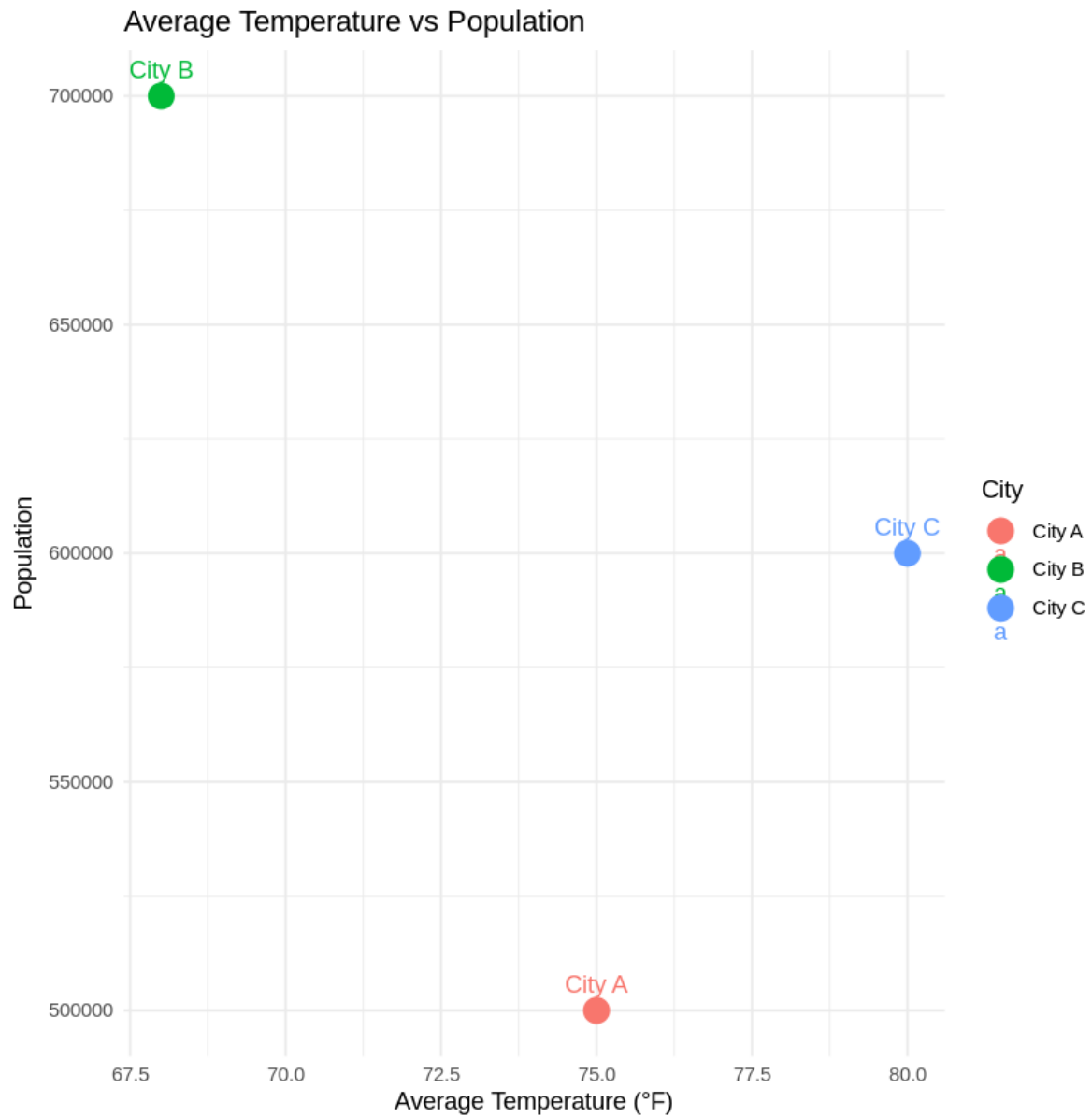
Program:

```
# Install package (Run once)
# install.packages("ggplot2")
library(ggplot2)
# Dataset
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)
# Scatter Plot
ggplot(geo_data,
  aes(x = Avg_Temperature,
    y = Population,
    color = City)) +
  geom_point(size = 5) +
  geom_text(aes(label = City), vjust = -1) +
  labs(title = "Average Temperature vs Population",
    x = "Average Temperature (°F)",
    y = "Population") +
  theme_minimal()
# Insights
cat("Insights:\n")
cat("- City B has the highest population (700,000).\n")
cat("- City C has the highest average temperature (80°F).\n")
cat("- No strong relationship can be determined with only three cities.\n")
```

Output:

Insights:

- City B has the highest population (700,000).
- City C has the highest average temperature (80°F).
- No strong relationship can be determined with only three cities.



3. Build a table to display the geographic data for each city. Label the table elements.

Program:

```
# Install packages (Run once)
# install.packages("gridExtra")
library(gridExtra)
library(grid)

# Dataset
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)

# Display Table
table_plot <- tableGrob(
  geo_data,
  rows = NULL,
  theme = ttheme_minimal()
)

grid.newpage()
grid.draw(table_plot)
```

Output:

City	Population	Avg_Temperature	Elevation
City A	5e+05	75	1000
City B	7e+05	68	800
City C	6e+05	80	1200

Set-14: Survey Results

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

Program:

```
# Install package (Run once)
# install.packages("ggplot2")
library(ggplot2)

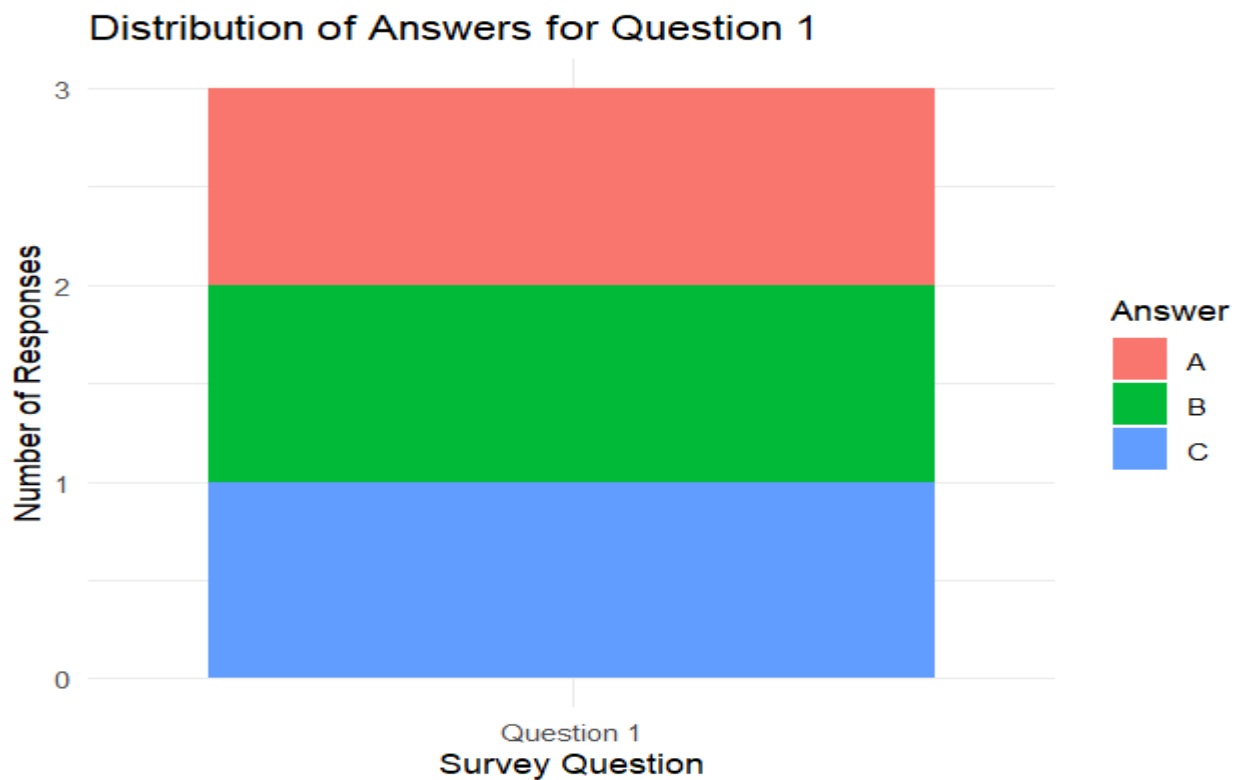
# Dataset
survey_data <- data.frame(
  Respondent = c(1, 2, 3),
  check.names = FALSE
)
survey_data$`Question 1` <- c("A", "B", "C")
survey_data$`Question 2` <- c("B", "A", "A")
survey_data$`Question 3` <- c("C", "D", "B")

# Count responses for Question 1
q1_counts <- as.data.frame(table(survey_data$`Question 1`))

# Rename columns
colnames(q1_counts) <- c("Answer", "Count")

# Stacked Bar Chart
ggplot(q1_counts,
  aes(x = "Question 1",
    y = Count,
    fill = Answer)) +
  geom_bar(stat = "identity") +
  labs(title = "Distribution of Answers for Question 1",
    x = "Survey Question",
    y = "Number of Responses",
    fill = "Answer") +
  theme_minimal()
```

Output:



2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

Program:

```
# Install package (Run once)
# install.packages("fmsb")

library(fmsb)

# Create dataset
survey_data <- data.frame(
  Respondent = c(1,2,3),
  Q1 = c(1,2,3), # A=1, B=2, C=3
  Q2 = c(2,1,1), # B=2, A=1, A=1
  Q3 = c(3,4,2)  # C=3, D=4, B=2
)

# Add maximum and minimum values
radar_data <- rbind(
  c(4,4,4), # Maximum
```

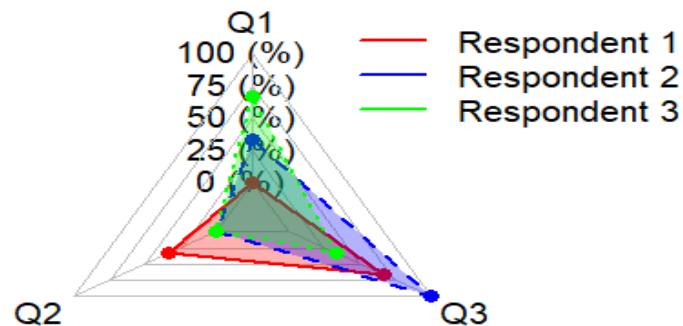
```

    c(1,1,1), # Minimum
    survey_data[,2:4]
)
# Draw radar chart
radarchart(
  radar_data,
  axistype = 1,
  pcol = c("red","blue","green"),
  pfc1 = c(rgb(1,0,0,0.3),
           rgb(0,0,1,0.3),
           rgb(0,1,0,0.3)),
  plwd = 2,
  cglcol = "grey",
  cglty = 1,
  axislabcol = "black",
  vlce = 1
)
# Add title
title("Overall Pattern of Survey Responses")
# Add legend
legend(
  "topright",
  legend = c("Respondent 1","Respondent 2","Respondent 3"),
  col = c("red","blue","green"),
  lty = 1,
  lwd = 2,
  bty = "n"
)

```


Output:

Overall Pattern of Survey Responses



3. Build a table to show the survey response data for each respondent. Label the table elements.

Program:

```
# Install packages (Run once)
# install.packages("gridExtra")
library(gridExtra)
library(grid)
# Dataset
survey_data <- data.frame(
  Respondent = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
# Display Table
table_plot <- tableGrob(
  survey_data,
```

```

rows = NULL,
theme = ttheme_minimal()
)
grid.newpage()
grid.draw(table_plot)

```

Output:

Respondent	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Set-15: Product Sales Analysis

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Program:

```

# Install packages (Run once)
# install.packages("ggplot2")
# install.packages("tidyr")
library(ggplot2)
library(tidyr)
# Dataset
sales_data <- data.frame(
  check.names = FALSE,
  "Product ID" = c(1,2,3),
  "Product Name" = c("Product A","Product B","Product C"),
  "January Sales" = c(2000,1500,1200),

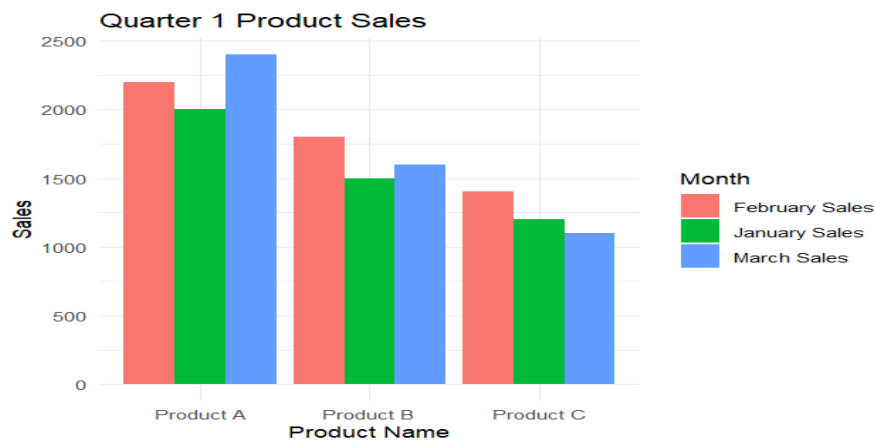
```

```

"February Sales" = c(2200,1800,1400),
"March Sales" = c(2400,1600,1100)
)
# Convert to long format
sales_long <- pivot_longer(
  sales_data,
  cols = c("January Sales", "February Sales", "March Sales"),
  names_to = "Month",
  values_to = "Sales"
)
# Grouped Bar Chart
ggplot(sales_long,
  aes(x = `Product Name`,
    y = Sales,
    fill = Month)) +
  geom_bar(stat = "identity",
    position = "dodge") +
  labs(title = "Quarter 1 Product Sales",
    x = "Product Name",
    y = "Sales",
    fill = "Month") +
  theme_minimal()

```

Output:



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Program:

```
# Install packages (Run once)
# install.packages("ggplot2")
# install.packages("tidyr")

library(ggplot2)
library(tidyr)

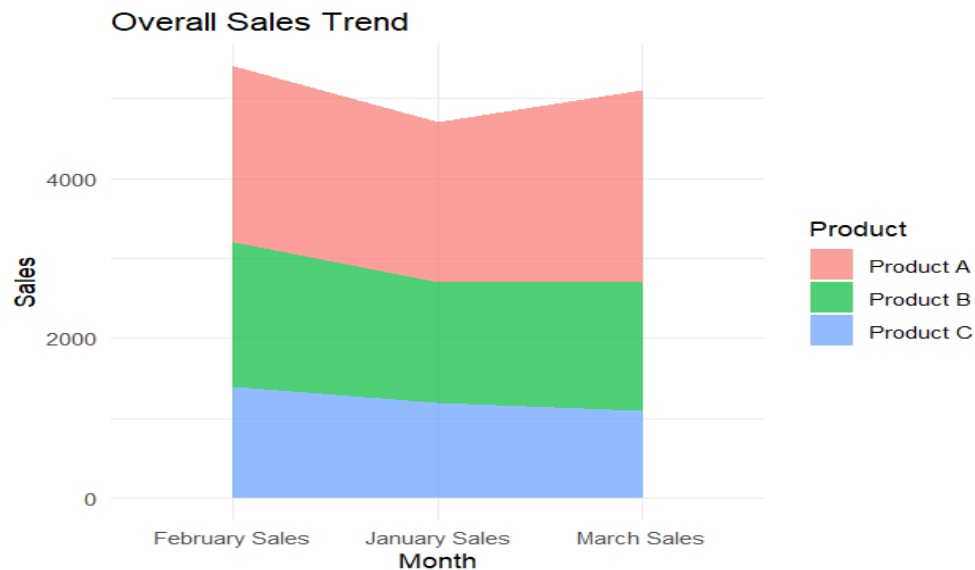
# Dataset
sales_data <- data.frame(
  check.names = FALSE,
  "Product ID" = c(1,2,3),
  "Product Name" = c("Product A","Product B","Product C"),
  "January Sales" = c(2000,1500,1200),
  "February Sales" = c(2200,1800,1400),
  "March Sales" = c(2400,1600,1100)
)

# Convert to long format
sales_long <- pivot_longer(
  sales_data,
  cols = c("January Sales","February Sales","March Sales"),
  names_to = "Month",
  values_to = "Sales"
)

# Stacked Area Chart
ggplot(sales_long,
  aes(x = Month,
    y = Sales,
    fill = `Product Name`,
    group = `Product Name`)) +
  geom_area(alpha = 0.7) +
```

```
labs(title = "Overall Sales Trend",
     x = "Month",
     y = "Sales",
     fill = "Product") +
theme_minimal()
```

Output:



3. Build a table to show the monthly sales data for each product. Label the table elements.

Program:

```
library(gridExtra)
library(grid)
# Dataset
sales_data <- data.frame(
  check.names = FALSE,
  "Product ID" = c(1,2,3),
  "Product Name" = c("Product A","Product B","Product C"),
  "January Sales" = c(2000,1500,1200),
  "February Sales" = c(2200,1800,1400),
  "March Sales" = c(2400,1600,1100)
)
# Create table
```

```

table_plot <- tableGrob(
  sales_data,
  rows = NULL,
  theme = ttheme_minimal()
)
# Display table
grid.newpage()
grid.draw(table_plot)

```

duct ID	Product Name	January Sales	February Sales	March Sa
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Output:

Set-16: Customer Demographics Analysis

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

Program:

```

# Install package (Run once)
# install.packages("ggplot2")
library(ggplot2)
# Dataset
customer_data <- data.frame(
  check.names = FALSE,

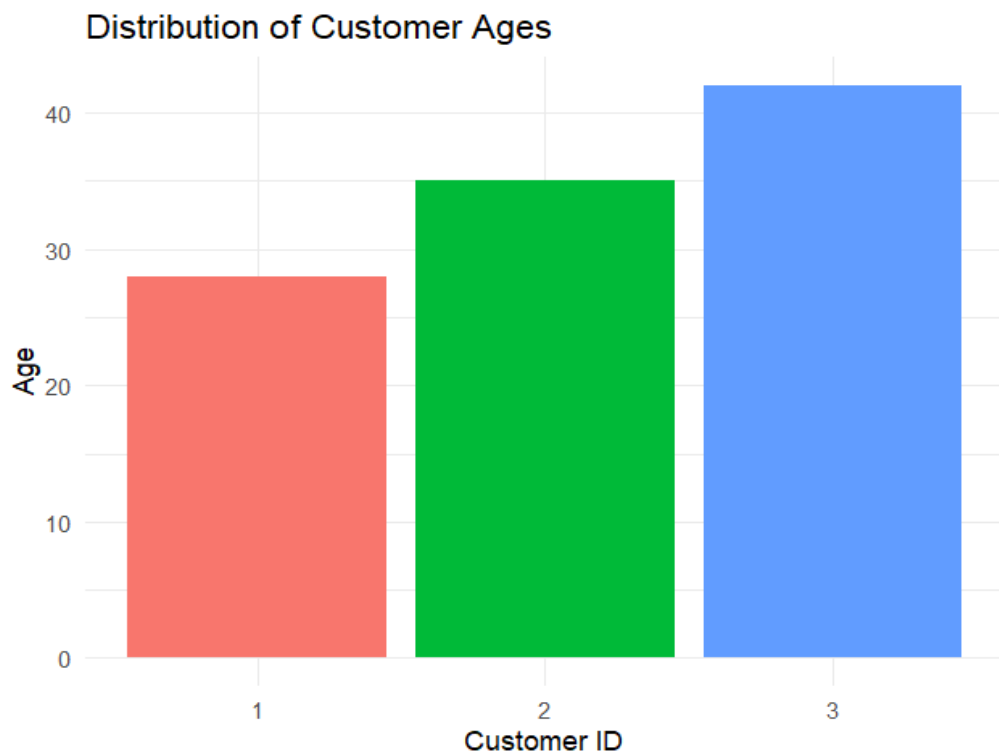
```

```
"Customer ID" = c(1,2,3),
"Age" = c(28,35,42),
"Gender" = c("Female","Male","Female"),
"Income (in $)" = c(50000,60000,75000)
)
```

```
# Bar Chart
```

```
ggplot(customer_data,
  aes(x = factor(`Customer ID`),
    y = Age,
    fill = factor(`Customer ID`))) +
  geom_bar(stat = "identity") +
  labs(title = "Distribution of Customer Ages",
    x = "Customer ID",
    y = "Age") +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:



2. Generate a pie chart to represent the distribution of customers by gender.

Program:

```
# Install package (Run once)
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Dataset
```

```
customer_data <- data.frame(
```

```
  check.names = FALSE,
```

```
  "Customer ID" = c(1,2,3),
```

```
  "Age" = c(28,35,42),
```

```
  "Gender" = c("Female","Male","Female"),
```

```
  "Income (in $)" = c(50000,60000,75000)
```

```
)
```

```
# Count Gender
```

```
gender_count <- as.data.frame(table(customer_data$Gender))
```

```
colnames(gender_count) <- c("Gender","Count")
```

```
# Pie Chart
```

```
ggplot(gender_count,
```

```
  aes(x = "", y = Count, fill = Gender)) +
```

```
geom_bar(stat = "identity", width = 1) +
```

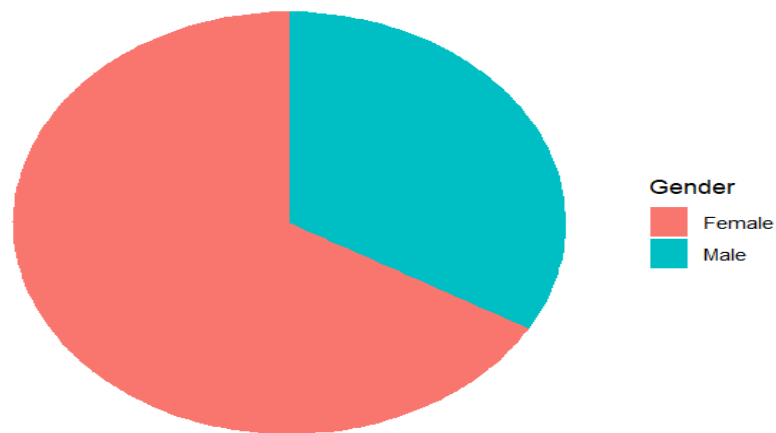
```
coord_polar("y") +
```

```
labs(title = "Distribution of Customers by Gender") +
```

```
theme_void()
```


Output:

Distribution of Customers by Gender



3. Build a table to show the demographic information for each customer. Label the table elements.

Program:

```
# Install packages (Run once)
# install.packages("gridExtra")
library(gridExtra)
library(grid)

# Dataset
customer_data <- data.frame(
  check.names = FALSE,
  "Customer ID" = c(1,2,3),
  "Age" = c(28,35,42),
  "Gender" = c("Female","Male","Female"),
  "Income (in $)" = c(50000,60000,75000)
)

# Display Table
table_plot <- tableGrob(
  customer_data,
  rows = NULL,
  theme = ttheme_minimal()
```

```
)  
grid.newpage()  
grid.draw(table_plot)
```

Output:

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Set-17: Employee Performance Analysis

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Program:

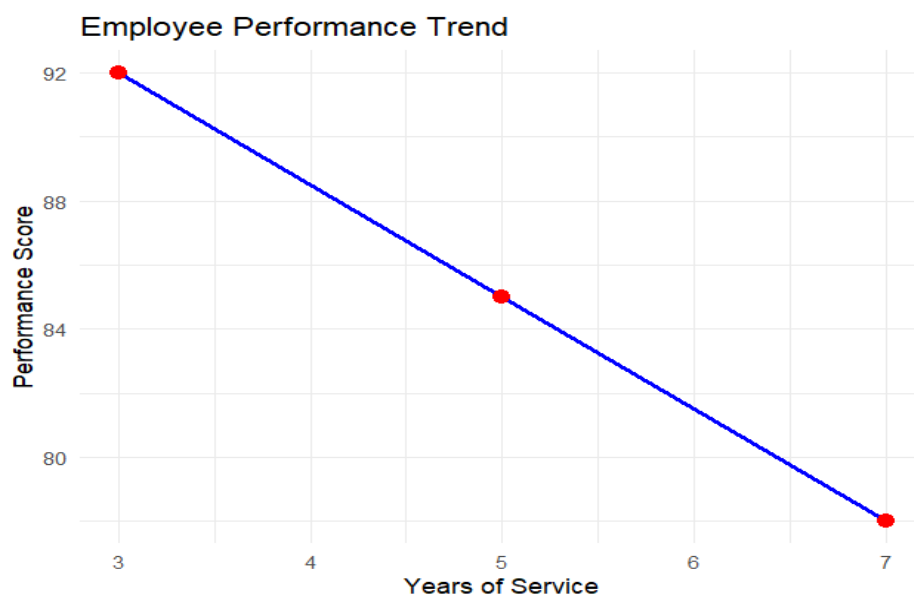
```
# Install package (Run once)  
# install.packages("ggplot2")  
library(ggplot2)  
# Dataset  
employee_data <- data.frame(  
  check.names = FALSE,  
  "Employee ID" = c(1,2,3),  
  "Department" = c("Sales","HR","Marketing"),  
  "Years of Service" = c(5,3,7),  
  "Performance Score" = c(85,92,78)  
)  
# Line Chart  
ggplot(employee_data,  
  aes(x = `Years of Service`,  
      y = `Performance Score`,
```

```

    group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score") +
  theme_minimal()

```

Output:



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Program:

```

# Install package (Run once)
# install.packages("ggplot2")
library(ggplot2)

# Dataset
employee_data <- data.frame(
  check.names = FALSE,
  "Employee ID" = c(1,2,3),
  "Department" = c("Sales","HR","Marketing"),

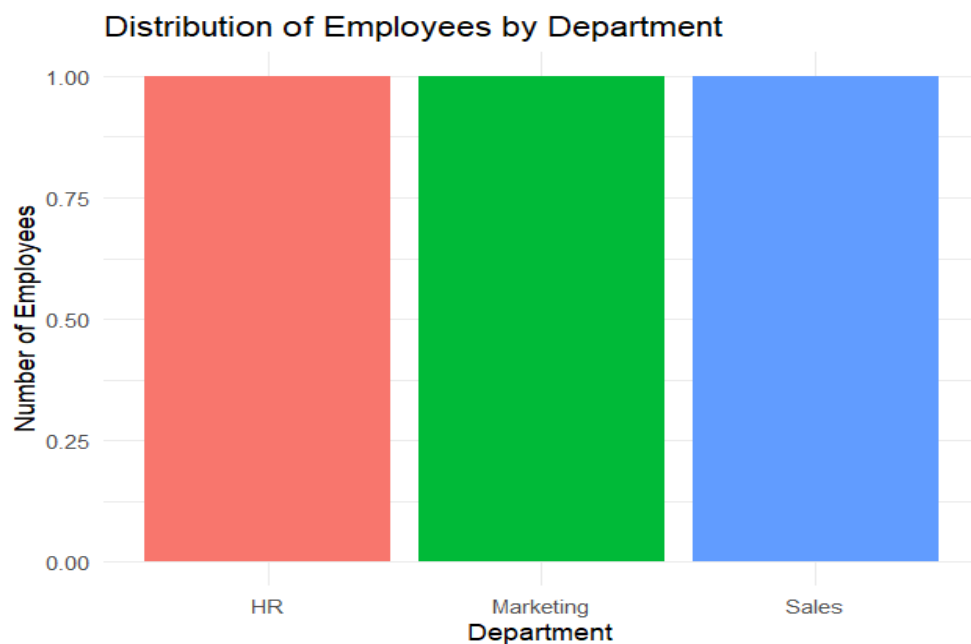
```

```

"Years of Service" = c(5,3,7),
"Performance Score" = c(85,92,78)
)
# Bar Chart
ggplot(employee_data,
  aes(x = Department,
    fill = Department)) +
  geom_bar() +
  labs(title = "Distribution of Employees by Department",
    x = "Department",
    y = "Number of Employees") +
  theme_minimal() +
  theme(legend.position = "none")

```

Output:



3. Build a table to display the performance data for each employee. Label the table elements.

Program:

```

# Install package (Run once)
# install.packages("gridExtra")
library(gridExtra)

```

```

library(grid)

# Dataset

employee_data <- data.frame(
  check.names = FALSE,
  "Employee ID" = c(1,2,3),
  "Department" = c("Sales", "HR", "Marketing"),
  "Years of Service" = c(5,3,7),
  "Performance Score" = c(85,92,78)
)

# Create Table

table_plot <- tableGrob(
  employee_data,
  rows = NULL,
  theme = ttheme_minimal()
)

# Display Table

grid.newpage()
grid.draw(table_plot)

```

Output:

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78